
A wintertime mission to the deep, convective, sinking ocean water in the Irminger Sea

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The Irminger Sea in the winter, west of Greenland, is a place with some of the harshest and unforgiving conditions in the world. A place where you will find waves the average size of three-story buildings, storms, freezing temperatures and extreme winds. Where the formation of cold convective water masses can sink hundreds or thousands of meters, driving global ocean circulation. A perfect place to go out on an expedition, right?

That is exactly what we did, all aboard the Anna Weber-van Bosse research vessel from the NIOZ institute, we set out from the port of Reykjavik. For any second thoughts, it was now too late, as the port behind us was shrinking in the distance. Thoughts of historic pioneering explorers flash through the mind, even more amplified as the first storms and icebergs appear on the horizon. Cliché or not, the Titanic and the adventure of Ernest Shackleton come to mind when you lay eyes upon what we captured below, in Figure 1. But why go there at all?

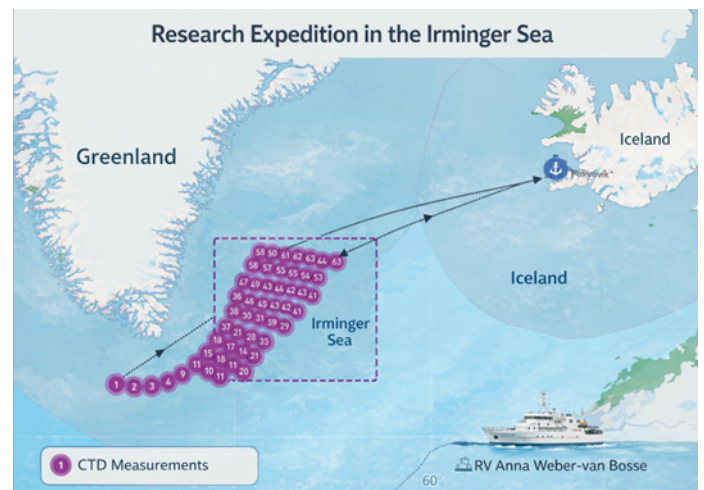


Figure 1 Harsh conditions on the Anna Weber-van Bosse as she conquers the storm. On the right the trajectory we followed with the rectangle depicting the research area.

The answer lies in a process called deep convection in which surface water sinks to depths exceeding 700 meters. This occurs when water becomes sufficiently dense, primarily due to cooling and increased salinity, allowing it to sink through the water column. In such cases, the density remains nearly uniform with depth, enabling vertical mixing, (Piron, 2016). Especially in intense winters, extreme winds cause extreme surface cooling, a process called atmospheric

forcing. This cooling increases the density of surface waters and triggers deep convection. This process plays a crucial role in the global overturning circulation.

In our expedition, CTD (Conductivity, Temperature, Depth) measurements were taken east of Greenland. These measurements allow us to determine the density of the water column. The 63 CTD measurements were deployed to a depth of 1500 m, one at each station. Our objective was to identify regions where density remains constant over at least 700 meters, which would be classified as deep convection. In addition to CTDs, we deployed Argo floats at stations 1 through 6, which are non-stationary measurement devices that can track ocean currents. In our case, we hoped it would follow the path where deep convection occurs.

When preparing such an expedition, you attempt to best predict where exactly you want to take measurements. Even with all the preparation in the world, luck will always be an ingredient that is required to actually measure the desired effect. In this expedition, this was no different. We were not off to a good start. The Argo floats we deployed drifted away in the opposite direction from where we wanted to measure, rendering the data useless for identifying deep convection. Our only hope was settled on the CTD measurements...

After two weeks at sea, we returned safely to Reykjavik. For the crew, this marked the end of a demanding expedition, because we went to the Irminger Sea in winter and survived. For us as oceanographers, however, the real work had only just begun. As soon as the data became available, we began analyzing the temperature and salinity profiles, searching for signs of deep convection.

The first 38 CTD stations revealed little evidence of deep convection. Then, at stations 39 and 44, something changed. At these stations, the density was remarkably uniform up to a depth of 900 meters, with a value of $\rho = 1027.70 \text{ kgm}^{-3}$.

When investigating this area more closely, we spotted something strange in the salinity profile at stations 39 and 44. Around 700 meters depth, a slightly fresher layer was present, a distinct water mass with different characteristics compared to its surroundings. At the same depth, the temperature decreased sharply. Together, these changes pointed to the presence of a distinct water mass hidden within the convective region. The salinity structure is shown in Figure 2.

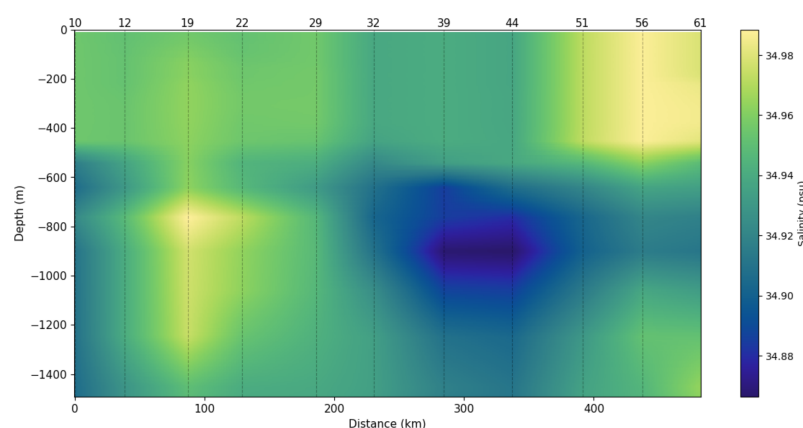


Figure 2 The salinity profile plotted with depth, between station 39 and 44 the relatively freshwater blob is easily observed. This is where deep convection occurs from the surface up to 900 meters.

Now the question remains, this fresh water blob, where does it come from and why is it there? This is important as it actually enables deep convection. From a modelling paper by Fried

(2025), the same freshwater blob has been found. This area has been identified as Labrador Sea Water (LSW), a water mass assumed to be formed in the Labrador Sea (west of Greenland).

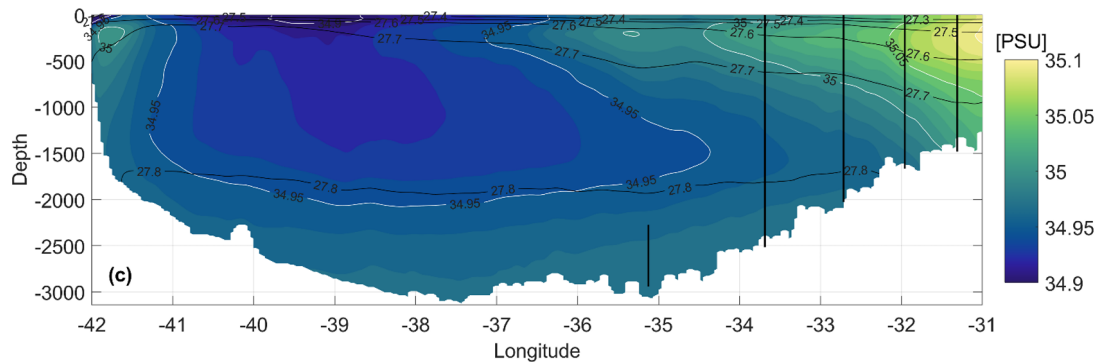


Figure 3 The salinity profile in the Irminger Sea as modelled by Fried et al. 2025. Between -38° and -40° the same relatively fresh water blob occurs. This water mass is identified as Labrador Sea Water.

Other studies have investigated why water with LSW characteristics can also be found in the Irminger Sea, and it is argued that this water mass can even be locally formed in the Irminger Sea. By boundary currents, this water from the Labrador Sea can flow into the Irminger Sea. Labrador Sea Water forms during intense winter convection, when the surface water cools it becomes denser and sinks to intermediate depths, (Yashayaev, 2024). This study also shows that the formation of Labrador Sea Water varies strongly from year to year, depending on winter conditions. Such water masses are typically formed in the Labrador Sea, but observations show that they do not remain confined there. Instead, they spread throughout the subpolar North Atlantic, (Chafik, 2022).

These distinct water masses do not only enable deep convection, and in turn can play a key role in the global ocean circulation. Understanding the origin and under what circumstances these can form is of great importance, especially under warming conditions, allowing us to predict how the ocean will circulate in the future. Yet these hidden layers of the ocean are not only relevant for climate science. In some cases, this creates so-called shadow zones, regions where sound waves bend in such a way that objects become difficult to detect. Submarines can exploit these zones to remain hidden, using the structure of the ocean itself as a natural form of camouflage (Gilli, 2024).

AI Disclosure

ChatGPT:

- Generating the images displayed in Figure 1. The following prompts were used:
 - “Generate me a similar picture, this is of the Irminger Sea. It represents harsh conditions, big waves, include ice bergs. Our research vessel is the RV Anna Weber-van Bosse of the NIOZ institute. It needs to be realistic.”
 - “Generate a picture of a map to show where our expedition took place. It was on the west of Greenland in the Irminger Sea. The purple data points is where we took CTD-measurements. I suggest marking that area by a rectangle or something. For the reader, it's mainly about where geographically we went to and which area we investigated.”
- To help improve the structuring and sentencing of the text:
 - Here is the article, I want you to improve the text grammatically below Figure 1. If it needs a bit of restructuring that is fine. But keep the overall structure still.

References

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